

Exercise 2.4. Prove the following. For each, m and n are integers.

(a) If m and n are odd, then $5m - 3n$ is even.

(b) If m and n are even, then $3mn$ is divisible by 4.

Solution:

For both proofs, we use the definitions:

$$\text{Even integer} = 2k, \quad \text{Odd integer} = 2k + 1$$

where $k \in \mathbb{Z}$.

(a) If m and n are odd, then $5m - 3n$ is even

Proof

Since m and n are odd, we can write

$$m = 2a + 1$$

and

$$n = 2b + 1$$

for some integers a, b .

Substitute these into $5m - 3n$:

$$5m - 3n = 5(2a + 1) - 3(2b + 1)$$

Expand:

$$= 10a + 5 - 6b - 3$$

$$= 10a - 6b + 2$$

Factor out 2:

$$= 2(5a - 3b + 1)$$

Since $5a - 3b + 1$ is an integer, $5m - 3n$ has the form $2k$.

Therefore,

$5m - 3n$ is even.

(b) If m and n are even, then $3mn$ is divisible by 4

Proof

Since m and n are even, we can write

$$m = 2a$$

and

$$n = 2b$$

for some integers a, b .

Consider $3mn$:

$$3mn = 3(2a)(2b)$$

Multiply:

$$= 3(4ab)$$

$$= 12ab$$

Now factor out 4:

$$= 4(3ab)$$

Since $3ab$ is an integer,

$$3mn = 4k$$

for some integer k .

Therefore, $3mn$ is divisible by 4.

$$\boxed{4 \mid 3mn}$$